



## Product Overview

### AX5043: Ultra-low Power RF Transceiver, 27-1050 MHz

For complete documentation, see the [data sheet](#)

#### Product Description

The AX5043 is a true single chip, narrow-band, ultra-low-power ASK and FSK RF transceiver for the 27 MHz to 1050 MHz frequency bands. It offers the unique combination of ultra-low power consumption for transmit and receive operation combined with highest sensitivity and high selectivity. A link budget of 143 dB at 1 kbps is achieved, if the built-in forward error correction (FEC) is used this can be extended to 146 dB without extra external components. The AX5043 works perfectly down to 1 kbps in a 6.25 kHz channel.

An integrated voltage regulation system allows the direct use of batteries and short start-up times enable time- and energy-efficient protocols. Transmit power and receive characteristics do not change over the supply voltage range of 1.8 - 3.6 V.

The AX5043 supports FSK, MSK, 4-FSK, GFSK, GMSK, AFSK and ASK modulations. In transmit mode, all modulations are shaped. For FSK gaussian filters with BT=0.3 or BT=0.5 are available to meet the most stringent regulatory requirements. Power ramping can be configured without restrictions. Maximum output power level is 16 dBm.

| Features                                                                                                                                                                                                                                                                                                          | Benefits                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>Carrier frequency range 27 - 1050 MHz</li><li>Ultra-low power</li><li>Constant TX output power over the full VDD range of 1.8 - 3.6 V</li><li>Fully autonomous lowest power wake-on-radio mode (500 nA)</li><li>Packet RX/TX without microcontroller intervention</li></ul> | <ul style="list-style-type: none"><li>Wide frequency range to work with</li><li>Achieves low power capability without sacrificing link budget or selectivity</li><li>Transmit power and receive characteristics do not change over the supply voltage range</li><li>Autonomously handles wake-on-radio cycling</li><li>Supports a wide range of packet formats that are handled without microcontroller intervention.</li></ul> |

#### Applications

- IOT (Internet of Things)
- Automated Meter Reading (AMR)
- Security Applications
- Building Automation
- Wireless Networks

## Part Electrical Specifications

| Product       | Compliance             | Status | Data Transmission Standard                                      | Frequency Band (MHz) | Carrier Frequency (MHz) | Package Type |
|---------------|------------------------|--------|-----------------------------------------------------------------|----------------------|-------------------------|--------------|
| AX5043-1-TA05 | Pb-free<br>Halide free | Active | GFSK<br>MSK<br>FSK<br>FM<br>AFSK<br>GMSK<br>4-FSK<br>ASK<br>PSK | 27-1050              |                         | QFN-28       |
| AX5043-1-TW30 | Pb-free<br>Halide free | Active | FM<br>GMSK<br>ASK<br>GFSK<br>PSK<br>MSK<br>FSK<br>4-FSK<br>AFSK | 27-1050              |                         | QFN-28       |

For more information please contact your local sales support at [www.onsemi.com](http://www.onsemi.com)

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